

Gemini Near Infra-Red Imager and Wavefront Sensor



Near IR Imager (NIRI) and On-Instrument Wavefront Sensor (OIWFS)—Implementation Notes and Record Reference

Hubert Yamada

niri_h ty_011 / 03 / 1999-10-23

This report contains a reference manual for several custom records and capfast symbols which are central to the implementation of The Gemini Near Infra-Red Imager and On Instrument Wavefront Sensor.

1 Introduction

This report presents a brief overview of the implementation of the control software for the Near Infra-Red Imager and On Instrument Wavefront Sensor. It also contains reference manuals for several custom EPICS records and capfast symbols.

1.1 Purpose of the document

This report provides a description of the interfaces and behavior of various NIRI software components.

The intended audience of this document is:

- the maintainers of the NIRI software;
- the reviewers of the NIRI system;
- the reviewers of the NIRI software;
- anyone who may wish to understand the details of NIRI.

2 Overview

The NIRI top-level directory (niri_VXYY, where XYY is a release number in the form major release, minor release) is made up of the following subdirectories

LIB. Shared utilities and schematics

CICSCC. NIRI CC code, adapted from the CICS

CICSWFS. NIRI WFS code, adapted from the CICS

ENG. NIRI mechanisms, WFS mechanisms, and NIRI environment control

DC. The GNAAC DC (Disabled)

Standalone subsystems of this system can be built by modifying the appropriate Makefiles and startup scripts in the root directory to compile only the subdirectories containing the desired subsystem, the LIB subdirectory, and the ENG subdirectory. For example, a stand-alone wavefront sensor can be built by including just the “LIB”, “CICSWFS”, and “ENG” directory. The ENG subdirectory is designed to build a product which is capable of functioning by itself.

The NIRI top-level directory, the LIB directory, and the CICSCC directories are adapted from the corresponding CICS directories, with the following changes:

- The niri_VXYY/ascii directory includes files from the ENG/ascii directory.
- The CICSCC directory has additional mechanisms, to simulate the entire NIRI instrument
- The channel-access links to the CICS motor simulator has been replaced with channel-access links to the ENG databases.

In addition, the CICSWFS directory is essentially a copy of the CICSCC directory, customized with the NIRI components.

2.1 The ENG directory

Note: The ENG directory is designed to be run as a standalone package for low-level engineering of the niri mechanisms, or as a part of the complete niri system.

The ENG directory contains three independent databases for controlling different mechanisms: The CC database, the WFS database, and the environment-control database. It contains a single database, the GBL database, which provides low-level interlocks. It contains databases which may be loaded to simulate the corresponding subsystems. There are also some special records, device support, and other C code which are required to support the databases.

It is possible to simulate the NIRI mechanisms, the WFS mechanisms, and the NIRI environment controls by loading the appropriate simulation database(s), which are also included in the ENG directory. The top-level user interface should be have identically with the simulation or with the actual hardware. Some slight differences may be noted in the lower-level engineering interfaces.

2.1.1 The CC Database

The components-controller database is made up of several instances of the same capfast hierarchical symbol, the mech symbol.

2.1.2 The WFS Database

The wavefront-sensor database is structured identically to the CC database, with a different set of mechanisms.

2.1.3 The GBL Database

The global parameters database provides access to interlocks, to keep the system from moving when it is unsafe (e.g., the mechanisms are not at an equilibrium temperature) or unacceptable (e.g., during an observation) for mechanisms to move. It also restricts the number of mechanisms that are in motion. (The power supply is not able to provide power for all motors simultaneously.)

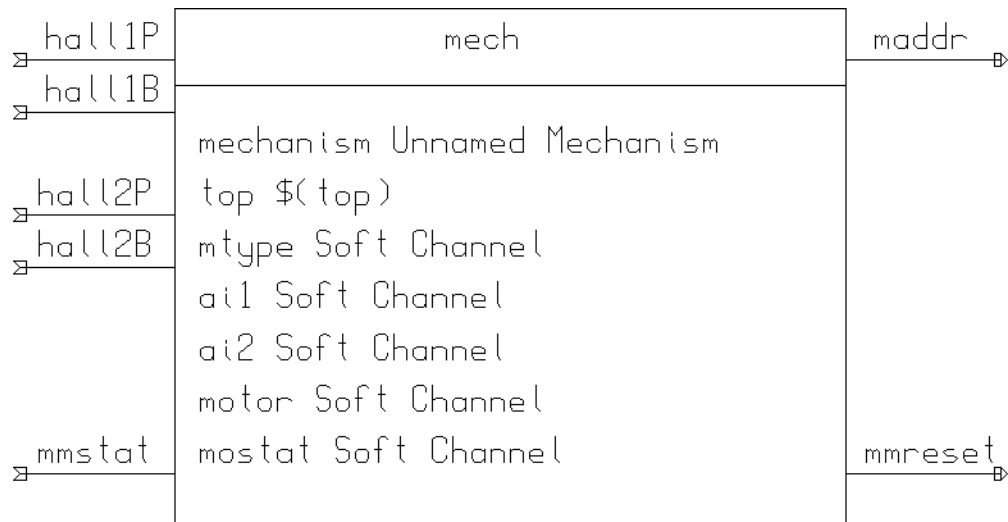
2.1.4 The ENV Database

The environment-control database is very simple, containing only a few records to monitor and to control the temperature sensors and temperature controllers.

3 The Mech Symbol and Diagram

All mechanical components are controlled by instances of a single capfast hierarchal symbol and the associated capfast diagram. Custom parameters are passed in via input and output ports (mostly hardware addresses) and via macro substitution.

3.0.1 The Mech Symbol



The following table describes the macro variables, inputs, and outputs of the mech hierarchal symbol.

TABLE 1. Mech.sym macro variables and ports

Macro/Port Name	Type	Value
hall1P	Input record hardware address	Primary Hall Effect sensor (wheel or main gear)
hall1B	Input record hardware address	Backup Hall Effect sensor (wheel or main gear)

TABLE 1. Mech.sym macro variables and ports

Macro/Port Name	Type	Value
hall2P	Input record hardware address	Primary Hall Effect sensor (linear sensor or unused)
hall2B	Input record hardware address	Secondary Hall Effect sensor (linear sensor or unused)
mmstat	Input record hardware address	Motor control module status (working or fault)
maddr	Motor control record hardware address	Motor controller board
mmreset	Output record hardware address	Motor control module reset (resets fault state)
mechanism	Macro variable	Text string (name of mechanism)
top	Macro variable	Prefix for generating record names
mttype	Macro variable	Type of motor controller (EPICS Device or Soft HS [simulator])
ai1	Macro variable	Type of input sensor (EPICS Device or Soft HS [simulator])
ai2	Macro variable	Type of input sensor (EPICS Device, Soft Channel [unused], or Soft HS [simulator])
motor	Macro variable	Type of motor (EPICS Device or Soft HS [simulator])
mostat	Macro variable	Type of motor status input (EPICS Device or Soft Channel [unused])

3.1 External interfaces (the NIRI/CICS interface)

This section describes the fields that the NIRI/CICS interface uses to control the mechanisms in the mech hierarchal symbol. These are the only fields that should be considered “public”¹. All other fields are to be considered a private to the implementation, and may be changed at any time.

TABLE 2. External Interfaces

Field	Action	Read or Write	Action	Comments
\$(top)MoveTo.VAL	Demand Position	Write	Move to new position	If an operation is already in progress, a new operation will fail.
\$(top)Op.VAL	Operation Code	Write	Perform operation.	See Table 3 on page 5 for a list of operation codes If an operation is already in progress, a new operation will fail.
\$(top)Stop.VAL	Stop	Write	Interrupt operation	Write a 1 to this field to interrupt an operation
\$(top)Busy.VAL	Busy	Read	Operation in progress?	Non-zero means operation in progress

1. The low-level engineering interface interacts with other fields; do not use it as an example of the correct way to interface with the mech hierarchal symbol.

The Mech Symbol and Diagram

TABLE 2. External Interfaces

Field	Action	Read or Write	Action	Comments
\$(top)Done.VAL	Done	Read	Status after operation	Non-zero means error
\$(top)Hallstep.MESS	Completion message	Read	Error/Status message after an operation	Cleared by the next operation
\$(top)Hallstep.DATM	Datumed?	Read or Write	Mechanism datumed?	Can be set or cleared externally or internally at any time. Set after successful datum, cleared if datum failed, cleared upon reboot. For informational purposes only; does not affect workings of record.
\$(top)Hallstep.HSWN	Hall effect sensor disabled?	Read	Hall-effect sensor disabled?	Can be set or cleared externally or internally at any time.
\$(top)Moving.VAL	Moving?	Read	Mechanism moving?	Non-zero means that it is moving. This is primarily for informational displays. Monitor \$(top)Busy.VAL instead, because the mechanism may briefly stop moving while an operation is still in progress.
\$(top)Mpos.VAL	Current Position?	Read	Mechanism current position	This field takes approximately 0.1 seconds to update. It is intended to be used for informational displays. For precise positioning, wait at least 0.1 second after Moving.VAL becomes 0 before reading this field.
\$(top)Speed.VAL	Desired speed?	Read	Mechanism speed	This is the desired speed after the mechanism has accelerated, <i>not</i> the current speed.

The following table describes all operation codes. *These codes might be out of date. Check values in hsConsts.h.*

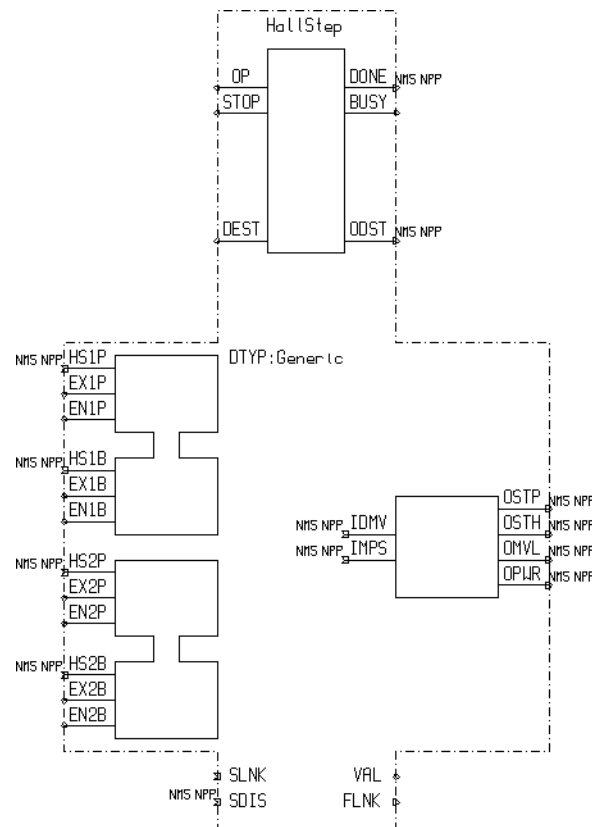
TABLE 3. Mech Hierarchical Symbol Operation codes

Code	Action	Public?	Comments
0	No Op	No	Reserved for internal use
1	Move	No	Use \$(top)MoveTo.VAL instead
2	Datum mechanism	No	For engineering. Normal programs should use Datum + Diagnose
3	Redatum mechanism	Yes	
4	Cycle mechanism	No	For engineering
5	Diagnose mechanism	Yes	
6	Datum + Diagnose	Yes	

4 The HallStep Record and Symbol

The “mech” diagram controls the hardware via the (custom) “hallStep” record. All commands to the hardware will pass through this record. (Hall effect sensor values may be read directly from the ai record.) Nothing should interface directly to the motor record, except for the records in the mech symbol.

In general, it should never be necessary to interface to the HallStep Record. Use the public fields in the mech symbol instead.



The following table describes the fields of the HallStep record.

TABLE 4. HallStep fields

Name	Summary	Description
VAL	Subr. return value	Error code returned if operation is rejected
OVAL	Old return value	
PREC	Display Precision	
STOP	Stop operation	Setting this field to non-zero cancels the current operation
OP	Operation	See Table 5 on page 9
DEST	Destination	Destination for Move operation

The HallStep Record and Symbol

TABLE 4. HallStep fields

Name	Summary	Description
OPST	Operation status	Text description of current action
MESS	Operation error msg	Text description of error or other important information
HOPR	High Operating Range	Upper limit of motion (Ignored during datum)
LOPR	Low Operating Range	Lower limit of motion (Ignored during datum)
RAW	Raw mode	Device dependant (e.g. ignore limits)
DONE	Operation error code	Output link, writes zero [no error] or non-zero [error] after operation completes
BUSY	Currently processing?	Output link, writes 1 when operation begins, 0 when operation ends
DSL P	“Datum slope P”	Exact meaning depends on device support (Wheel or Stage), used for datuming
DSL B	“Datum slope B”	Exact meaning depends on device support (Wheel or Stage), used for datuming
DOFP	“Datum offset, rough P”	Exact meaning depends on device support (Wheel or Stage), used for datuming
DOFB	“Datum offset, rough B”	Exact meaning depends on device support (Wheel or Stage), used for datuming
DBSH	“Datum backlash”	Exact meaning depends on device support (Wheel or Stage), used for datuming
DOF1	“Sensor offset, 1P-1B”	Exact meaning depends on device support (Wheel or Stage), used for datuming
WTHP	“Peak width 1P”	Estimated width of sensor peak (gear or wheel)
PKP	“Peak height 1P”	Estimated height of sensor peak (gear or wheel)
WTHB	“Peak width 1B”	Exact meaning depends on device support (Wheel or Stage)
PKB	“Peak height 1B”	Exact meaning depends on device support (Wheel or Stage)
CSTP	“Cycle stop”	Stop cycle test
CTOL	“Cycle tolerance”	Sensor error for automatic stop
CSET	“Cycle set values”	(Flag) On next cycle, set expected values for cycle test from observed value
CDSA	“Cycle A dest.”	First position for cycle test
CDSB	“Cycle B dest.”	Second position for cycle test
CDSC	“Cycle C dest.”	Third position for cycle test (normally 0)
E1PA	“Cycle A expect”	Expected primary (gear or wheel) sensor value at first test position
E1PB	“Cycle B expect”	Expected primary (gear or wheel) sensor value at second test position
E1PC	“Cycle C expect”	Expected primary (gear or wheel) sensor value at third test position
V1PA	“Cycle A read”	Actual primary (gear or wheel) sensor value at first test position
V1PB	“Cycle B read”	Actual primary (gear or wheel) sensor value at first test position
V1PC	“Cycle C read”	Actual primary (gear or wheel) sensor value at first test position
E1BA	“Cycle A expect”	Expected secondary (gear or wheel) sensor value at first test position
E1BB	“Cycle B expect”	Expected secondary (gear or wheel) sensor value at second test position
E1BC	“Cycle C expect”	Expected secondary (gear or wheel) sensor value at third test position
V1BA	“Cycle A read”	Actual secondary (gear or wheel) sensor value at first test position
V1BB	“Cycle B read”	Actual secondary (gear or wheel) sensor value at second test position
V1BC	“Cycle C read”	Actual secondary (gear or wheel) sensor value at third test position
E2PA	“Cycle A expect”	Expected primary (linear) sensor value at first test position
E2PB	“Cycle B expect”	Expected primary (linear) sensor value at second test position

The HallStep Record and Symbol

TABLE 4. HallStep fields

Name	Summary	Description
E2PC	“Cycle C expect”	Expected primary (linear) sensor value at third test position
V2PA	“Cycle A read”	Actual primary (linear) sensor value at first test position
V2PB	“Cycle B read”	Actual primary (linear) sensor value at second test position
V2PC	“Cycle C read”	Actual primary (linear) sensor value at third test position
E2BA	“Cycle A expect”	Expected secondary (linear) sensor value at first test position
E2BB	“Cycle B expect”	Expected secondary (linear) sensor value at second test position
E2BC	“Cycle C expect”	Expected secondary (linear) sensor value at third test position
V2BA	“Cycle A read”	Actual secondary (linear) sensor value at first test position
V2BB	“Cycle B read”	Actual secondary (linear) sensor value at second test position
V2BC	“Cycle C read”	Actual secondary (linear) sensor value at third test position
HSWN	“Warn HS”	Flag set when any hall effect sensor is disabled
HS1P	“Input Link HS1P”	Link to Hall-effect sensor input record
EX1P	“Expect HS1P”	Expected value of sensor after next move command
EN1P	“Enable HS1P”	Enable/Disable Hall-effect sensor
HS1B	“Input Link HS1B”	Link to Hall-effect sensor input record
EX1B	“Expect HS1B”	Expected value of sensor after next move command
EN1B	“Enable HS1B”	Enable/Disable Hall-effect sensor
HS2P	“Input Link HS2P”	Link to Hall-effect sensor input record
EX2P	“Expect HS2P”	Expected value of sensor after next move command
EN2P	“Enable HS2P”	Enable/Disable Hall-effect sensor
HS2B	“Input Link HS2B”	Link to Hall-effect sensor input record
EX2B	“Expect HS2B”	Expected value of sensor after next move command
EN2B	“Enable HS2B”	Enable/Disable Hall-effect sensor
IDMV ^a	“Motor finished moving”	Input link, used to monitor motion flag on motor-control record
IMPS ^a	“Motor Cur. position”	Input link, used to monitor motion flag on motor-control record
OMVL ^a	“Motor New position”	Output link, used to write new position on motor-control record
OSTH ^a	“Motor Set home”	Output link, used to trigger “set home” operation on motor-control record
OSTP ^a	“Motor Stop motion”	Output link, used to write “stop” on motor-control record
OPWR ^a	“Motor Power”	Output link, used to control power to on motor-control record
ODST	“Destination Link”	Output link, after a move, writes the current position

a. IDMV, IMPS, OMVL, OSTH, OSTP, and OPWR are very closely tied with the behavior of the motor record. If the motor record is modified, the behavior of these fields may need to change.

The following table describes all operation codes. *These codes might be out of date. Check values in `hsConsts.h`.*

TABLE 5.**HallStep Record Operation Codes**

Code	Action
0	No Op
1	Move
2	Datum mechanism
3	Redatum mechanism
4	Cycle mechanism
5	Diagnose mechanism
6	Datum + Diagnose

4.1 Record processing

This record is processed only when an operation code is written to the OP field.

4.1.1 When an operation code is processed:

If the record is processing another operation, the new operation fails, and the VAL field is set to a non-zero value.

If the operation code is invalid, the new operation fails and the VAL field is set to a non-zero value.

If the operation code is valid, the operation code is passed to the completion task, and the completion task is awakened.

Record processing completes, and 0 is returned in the VAL field.

4.1.2 When the completion task is given a new operation

The task writes a non-zero value to the BUSY link.

The task performs the desired operation (described below), reading sensor values from the sensor input records, writing control codes to the motor-control record. If there is an error, the task sets the MESS field to a descriptive message.

The task writes an error code (or 0 if there is no error) to the OPST field.

The task writes a 0 to the BUSY link.

For the move command:

The DEST field is read.

The hardware is moved to the new location.

The sensor values are compared to the expected sensor values (not yet implemented).

For the datum command:

The DATM field is set to 0

The hardware performs whatever operations are necessary to datum the hardware (see Section 3.1 on page 5 of [3]).

If the operation succeeded, the new datum is set.

If the operation succeeded, the DATM field is set to 1

For the redatum operation

The hardware performs whatever operations are necessary to datum the hardware (see Section 3.1 on page 5 of [3]).

The new datum is *not* set; the MESS field indicates how different the new datum is from the last datum

For the cycle operation

The hardware moves to the first test position (set by the CDSA, field. At the first test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the second test position (set by the CDSB, field.

At the second test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the third test position (set by the CDSC, field.

At the third test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The process begins again, starting with the first test position, until the operation fails, or until the CSTP field is set to a non-zero value.

For the diagnose operation

The hardware moves to the first test position (set by the CDSA, field.

At the first test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the second test position (set by the CDSB, field.

At the second test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the third test position (set by the CDSC, field.

At the third test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

For the datum + diagnose operation

The DATM field is set to 0

The hardware performs whatever operations are necessary to datum the hardware (see Section 3.1 on page 5 of [3]).

If the operation succeeded, the new datum is set.

The hardware moves to the first test position (set by the CDSA, field).

At the first test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the second test position (set by the CDSB, field).

At the second test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the third test position (set by the CDSC, field).

At the third test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

If the operation succeeded, the DATM field is set to 1

